

Null Foliations for Scattering Problems

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Outline

Intro to numerical conformal scattering

Scalar wave equation in Minkowski

Regge-Wheeler equation in Schwarzschild

Summary & outlook

Outline

Intro to numerical conformal scattering

- Intro of intro

- Numerical development of conformal scattering

Scalar wave equation in Minkowski

Regge-Wheeler equation in Schwarzschild

Summary & outlook

Linear scattering on Schwarzschild

The Regge–Wheeler–Zerilli equation in frequency domain, Schrödinger-like master equation

$$\mathcal{L}_{\ell\omega}^{\text{RW/Z}}\psi := \frac{d^2\psi}{dr_*^2} + (\omega^2 - V_\ell^{\text{RW/Z}})\psi = 0$$

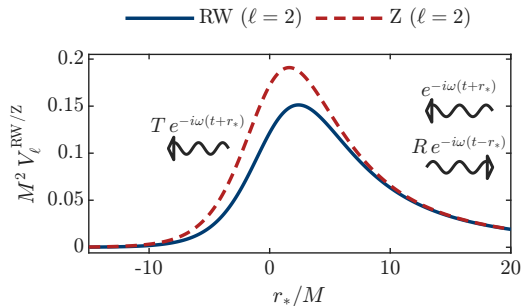
with tortoise $r_* = r + 2M \ln(r/2M - 1)$ and potential

$$V_\ell^{\text{RW/Z}} \longrightarrow \begin{cases} e^{r_*/(2M)}, & r_* \rightarrow -\infty, \\ \frac{\ell(\ell+1)}{r^2}, & r_* \rightarrow +\infty; \end{cases}$$

► (stationary) scattering boundary condition:

$$\begin{cases} e^{-i\omega r_*} + R e^{+i\omega r_*}, & r_* = +\infty \\ T e^{-i\omega r_*}, & r_* = -\infty \end{cases}$$

► Balance law: $|R|^2 + |T|^2 = 1$.



Nonlinear scattering on Schwarzschild (frequency domain)

Nonlinear scattering of a monochromatic wave [Cardoso, Redondo-Yuste, Sperhake, Tuncer, arXiv:2603.04501]

► Perturbation hierarchy

$$\psi = \epsilon \psi^{(1)} + \epsilon^2 \psi^{(2)} + \dots$$

► Linear drive: $\mathcal{L}_{\ell\Omega_d}^{\text{RW}} \psi_{\text{odd}}^{(1)} = 0$

► Quadratic response:

$$\mathcal{L}_{2\ell, 2\Omega_d}^{\text{Z}} \psi_{\text{even}}^{(2)} = S^{(2)} [\psi_{\text{odd}}^{(1)}, \psi_{\text{odd}}^{(1)}]$$

► Interesting nonlinear results: spectral broadening, focusing, and resonance near fundamental modes $\text{Re } \omega_{\ell 00}$.

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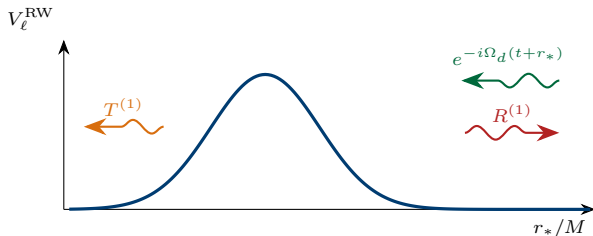
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linear scattering of the drive: odd RW channel, (ℓ, Ω_d)

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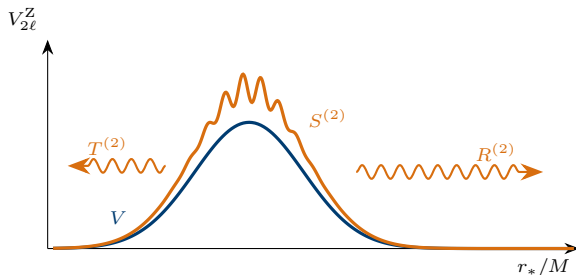
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quadratic response: even Zerilli channel $(2\ell, 2\Omega_d)$ — regularised source $S^{(2)}$ (wavy) with the same asymptotic behavior as V (smooth); no incident wave

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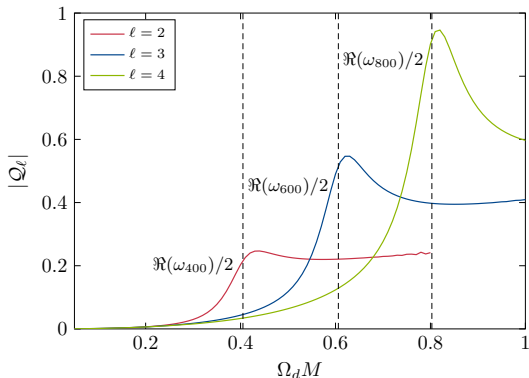
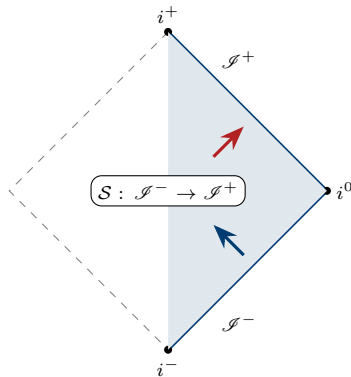


Fig. 3 of [Cardoso et al., arXiv:2603.04501]: $\mathcal{Q} \propto R^{(2)}/(R^{(1)})^2$ peaks near $2\Omega_d = \text{Re } \omega_{\ell 00}$ ($\ell = 2, 3, 4$).

Conformal scattering (time domain)

- ▶ A beautiful framework: the conformal method by Penrose.
- ▶ A Goursat (characteristic) problem: data on $\mathcal{I}^-$ evolve to data on $\mathcal{I}^+$.
- ▶ Obstacles: singular i^0 and $i^\pm$.
- ▶ It's an active research area in mathematical relativity, see [J.-P. Nicolas, Phil. Trans. R. Soc. A **382**, 20230035 (2024)] for a brief review.



Numerical implementation of Generalized Conformal Field Equations (GCFE)

- ▶ GCFE: a regular reformulation of the Einstein equations on a conformally rescaled spacetime — $\mathcal{I}$ becomes a computational boundary.
- ▶ A series of works: The non-linear perturbation of a black hole by gravitational waves [Frauendiener et al., CQG (2021–2025)]
- ★ Breakthrough: The first numerical simulations covering $\mathcal{I}^\pm$ of asymptotically flat spacetimes [Frauendiener, Stevens, Thwala, PRL (2025)].
- ▶ Necessary improvements:
 - ▶ to prescribe the initial data at $\mathcal{I}^-$
 - ▶ to include i^0 for a fully global evolution
- ▶ Fully global numerical evolutions of the linearised spin-2 equation [Barrar et al, arXiv:2608.01541]

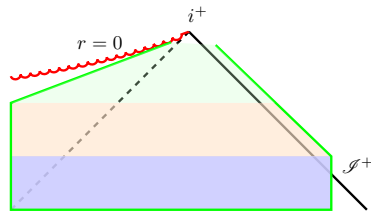


Fig. 1 of [Class. Quantum Grav. (2021)]: the computational domain evolves with the regridding procedure — blue: no regridding; orange: regridding outside $\mathcal{I}^+$; green: regridding also inside the black hole.

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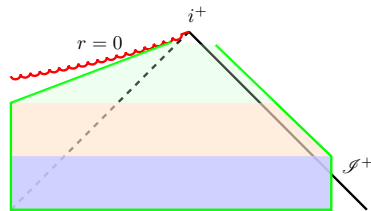


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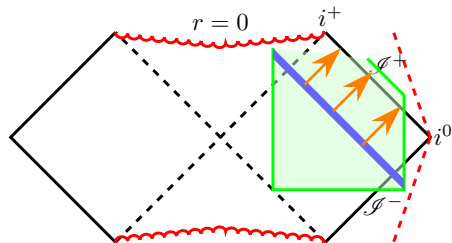


Fig. 1 of PRL (2025). Their **computational domain (green)** over the conformal diagram: **blue strip** — **ingoing GW** crossing $\mathcal{I}^-$; **orange arrows** — **outgoing radiation** from nonlinear back-reaction; **red dashed** — **hyperbolicity-loss surface** of the spin-2 subsystem.

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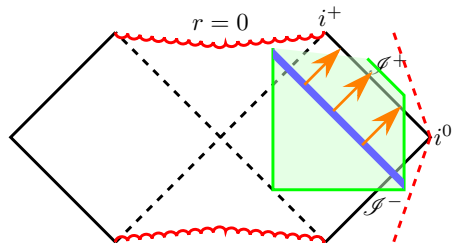


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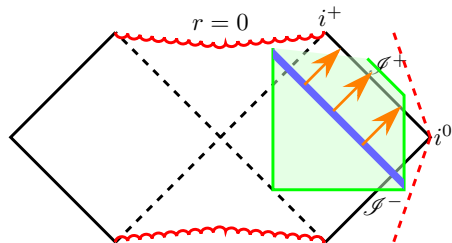


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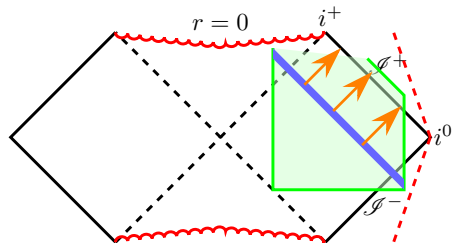
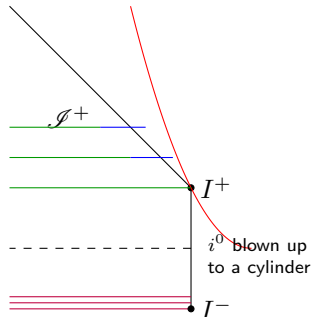


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A hybrid gauge of Minkowski spacetime in [Barrar et al., arXiv:2608.01541]

Hyperboloidal route to conformal scattering

- ▶ **Baptista, Vañó-Viñuales, and del Río**
[arXiv:2506.18610]: past and future hyperboloidal slices are connected through interpolation.

- ▶ Asymptotic radiation on $\mathcal{I}^\pm$ is directly accessible.
- ▶ However, there is a **causal gap** near i^0 .

- ▶ **Demirboğa and Zenginoğlu** [arXiv:2605.26028]: insert a central Penrose patch
 - ▶ interpolation $\rightarrow$ exact matching
 - ▶ to include the neighborhood of i^0 as part of the computational domain

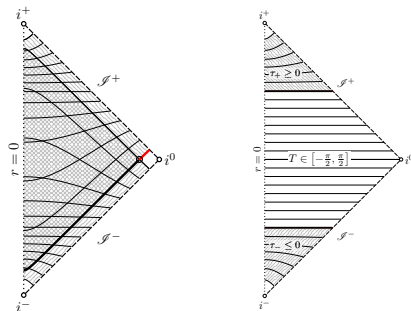


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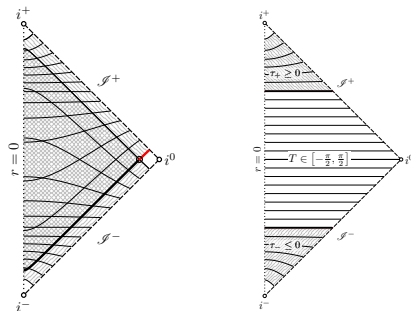


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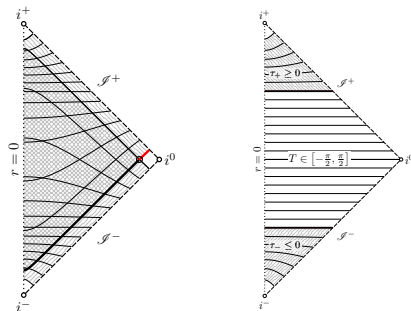


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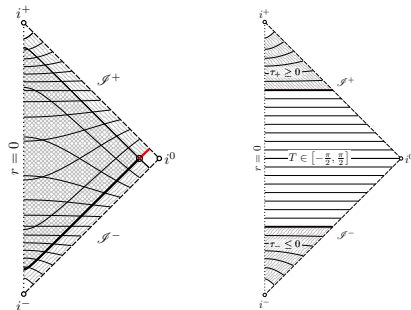


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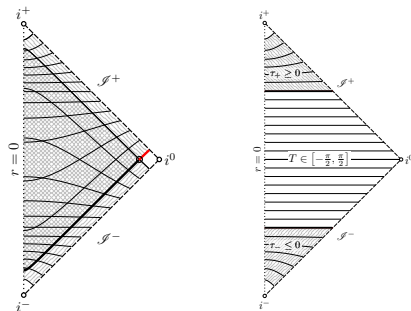


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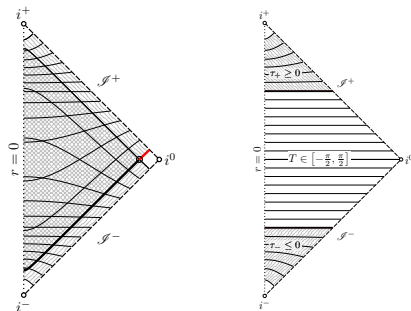


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This work: null foliations for conformal scattering

Null foliations: a unified (simple) numerical framework

- ▶ with radiation on $\mathcal{I}^\pm$ directly accessible
- ▶ to include i^0
- ▶ check of stability, convergence, and scattering quantities

Two companion works on characteristic evolution of conformal scattering

- ▶ 1. Scalar wave equation in Minkowski spacetime [arXiv:2608.15729]
- ▶ 2. Regge-Wheeler equation in Schwarzschild spacetime [ongoing]

Outline

Intro to numerical conformal scattering

Scalar wave equation in Minkowski

2.1 Geometric setup

2.2 Numerical stencils

2.3 Results

Regge-Wheeler equation in Schwarzschild

Summary & outlook

Compactified double-null coordinates

Minkowski spacetime in spherical coordinates $(t, r, \vartheta, \varphi)$

$$\eta = ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2$$

- ▶ Penrose coordinates (T, R) :

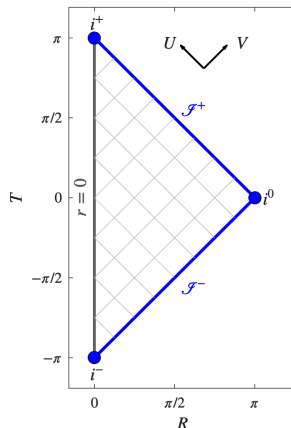
$$T = \arctan(t+r) + \arctan(t-r), \quad R = \arctan(t+r) - \arctan(t-r)$$

- ▶ Compactified double-null coordinates (U, V) :

$$U = T - R, \quad V = T + R$$

- ▶ Conformal boundary $(\Omega_P := 2 \cos \frac{U}{2} \cos \frac{V}{2} = 0)$:

- ▶ Timelike infinity $i^\pm$: $(T, R) = (\pm\pi, 0)$
- ▶ Null infinity: $\mathcal{I}^+$ ($V = \pi$), $\mathcal{I}^-$ ($U = -\pi$)
- ▶ Spatial infinity i^0 : $(T, R) = (0, \pi)$



Penrose diagram of Minkowski spacetime.

The scalar wave equation in the (U, V) coordinates

Consider $\square\phi = \mathcal{S}$, rescaled by $\psi = r\phi$

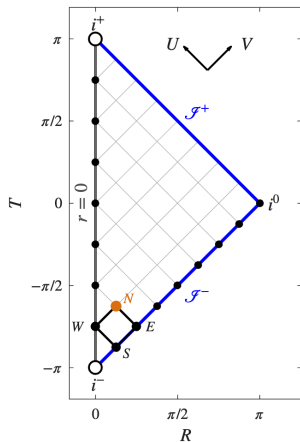
$$\left(\partial_U \partial_V - \frac{\Delta_\Omega}{4 \sin^2 R}\right) \psi = -\frac{r\mathcal{S}}{4\Omega_P^2}$$

After mode decomposition,

$$\partial_U \partial_V \psi_{\ell m} = -V_\ell \psi_{\ell m} - \frac{r\mathcal{S}_{\ell m}}{4\Omega_P^2}, \quad V_\ell = \frac{\ell(\ell+1)}{4 \sin^2 R}$$

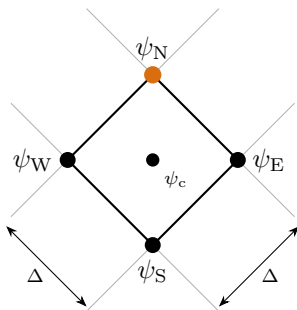
Characteristic initial-boundary value problem

- ▶ characteristic data on $\mathcal{I}^-$ (incl. i^0)
- ▶ regularity condition at $r = 0$: $\psi = r\phi = 0$



- prescribed data on $\mathcal{I}^-$ (incl. i^0) and $r = 0$;
○: $i^\pm$ unnecessary for our computation

Numerical stencils for characteristic integration



The cell $\mathcal{C}$ with known ψ_W , ψ_E and ψ_S , and unknown ψ_N .

Integrate $\partial_U \partial_V \psi = -S(U, V; \psi)$ over a cell $\mathcal{C}$:

$$\begin{aligned}\psi_N - \psi_E - \psi_W + \psi_S &= - \iint_{\mathcal{C}} S \, dU \, dV \\ &= -S(U_c, V_c; \psi_c) \Delta^2 - I_\epsilon,\end{aligned}$$

R.H.S. is approximated by **the midpoint value** $\times$ **the cell area** Δ^2

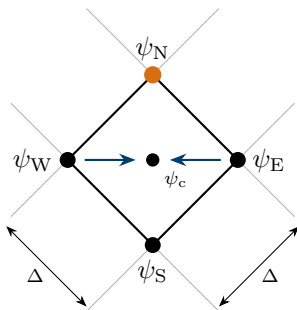
- ▶ Explicit stencil: $\psi_c \simeq \frac{\psi_E + \psi_W}{2}$
- ▶ Implicit stencil: $\psi_c \simeq \frac{\psi_N + \psi_S}{2}$

Convergence:

if S is smooth (C^2 at least), the truncation error $I_\epsilon \sim \partial^2 S \times O(\Delta^4)$, yielding a **second-order** method ($O(\Delta^2) = O(\Delta^4)/\Delta^2$).

However, this might fail at the conformal boundary $\Omega_P = 0$, thereby reducing the convergence order.

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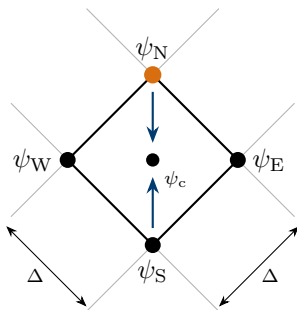
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- ▶ Explicit stencil: $\psi_c \simeq \frac{\psi_E + \psi_W}{2}$
- ▶ Implicit stencil: $\psi_c \simeq \frac{\psi_N + \psi_S}{2}$

Convergence:

if S is smooth (C^2 at least), the truncation error $I_\epsilon \sim \partial^2 S \times O(\Delta^4)$, yielding a **second-order** method ($O(\Delta^2) = O(\Delta^4)/\Delta^2$).

However, this might fail at the conformal boundary $\Omega_P = 0$, thereby reducing the convergence order.

Stability of the stencils: $|A| \leq 1$

Linear cases: $S = V_{\text{eff}}\psi$, $c = \Delta^2 V_{\text{eff}}(U_c, V_c)$

$$\psi_N = \mathbf{A}(\mathbf{c})(\psi_E + \psi_W) - \psi_S$$

$$\begin{cases} \text{Explicit : } A = (1 - c/2) \\ \text{Implicit : } A = 1/(1 + c/2) \end{cases}$$

An empirical rule sufficient for stability: $|A| \leq 1$

$$\begin{cases} \text{Explicit : } 0 \leq c \leq 4 \\ \text{Implicit : } c \geq 0 \quad (\text{always stable}) \end{cases}$$

The special case $\ell = 1$

- ▶ The implicit stencil gives first-order convergent outgoing radiation for $\ell = 1$. (Why?)
- ▶ An adaptive stencil for $\ell = 1$: $0 \leq c < 4$ choose explicit, $c \geq 4$ choose implicit.

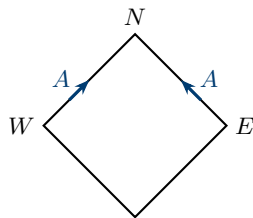


Illustration of the empirical rule for stability.

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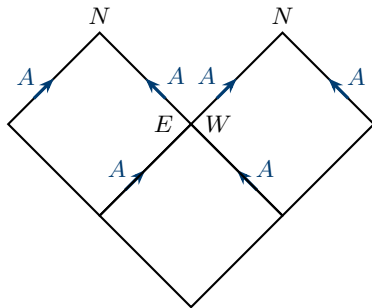


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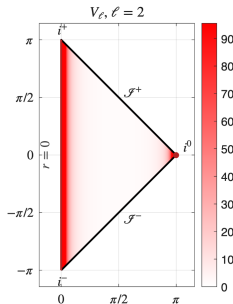
Physical scenarios for $\partial_U \partial_V \psi = -S$

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► **Free wave:** $V_{\text{eff}} = V_\ell = \frac{\ell(\ell+1)}{4 \sin^2 R}$

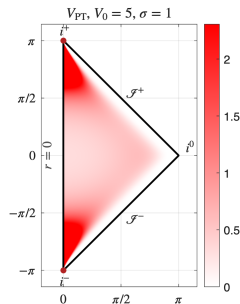
► **Pöschl–Teller:** $V_{\text{eff}} = V_\ell + V_{\text{PT}}$,

$$V_{\text{PT}} = \frac{V_0 \operatorname{sech}^2(r/\sigma)}{4\Omega_{\text{P}}^2}$$



V_ℓ **diverges** as $\ell(\ell+1)/\Delta^2$ at $R=0$ and i^0 .

$c_{\text{max}} = \ell(\ell+1)$. For $\ell = 1$, the explicit stencil is stable.



V_{PT} **diverges** in a direction-dependent way at $i^\pm$.
 Cut off the $i^\pm$ corner or use the adaptive stencil.

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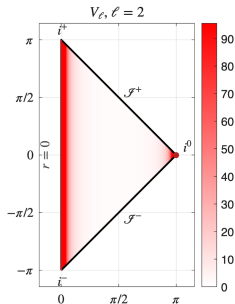
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Semilinear ($\square \phi = \lambda |\phi|^{n-1} \phi$, spherical):

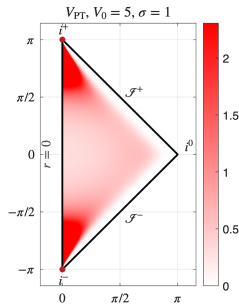
$$S = \frac{\Omega_{\text{P}}^{n-3}}{4 \sin^{n-1} R} \lambda |\psi|^{n-1} \psi, \quad \lambda > 0 \text{ (defocusing)}$$

The explicit stencil is numerically stable in this case.



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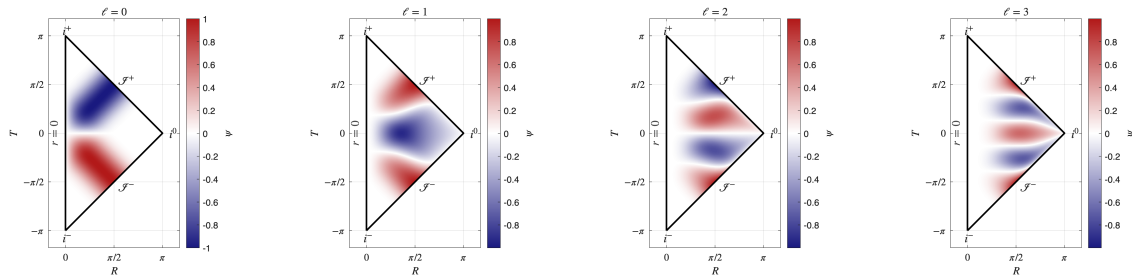
Results

Numerically compactly supported Goursat data on $\mathcal{I}^-$: modulated Gaussian

$$\mathcal{R}_- := \psi_{\text{inc}}(v) = A_0 \exp\left[-\frac{(v - v_0)^2}{2\sigma_w^2}\right] e^{-i\omega_0(v - v_0)},$$

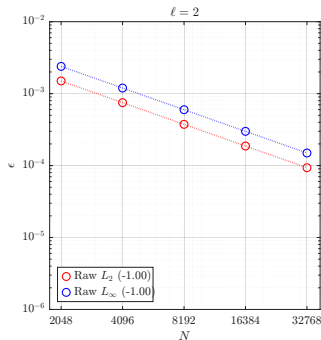
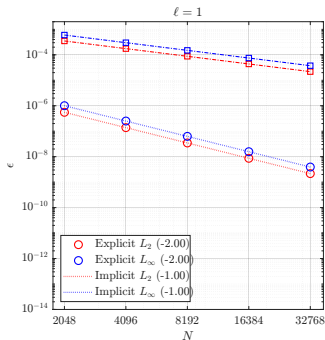
with physical advanced time $v = \tan \frac{V}{2}$.

Free propagation: (patterns with the correct parity)



For $\ell = 0$, the stencils reduce to the exact discrete integral, so the error is at roundoff level.

Convergence of outgoing radiation $\mathcal{R}_+ := -\psi|_{\mathcal{I}^+}$

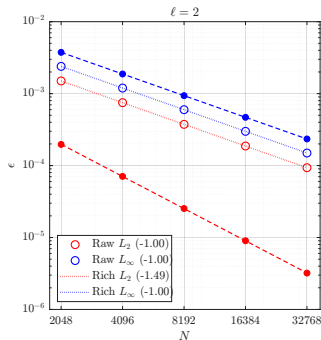
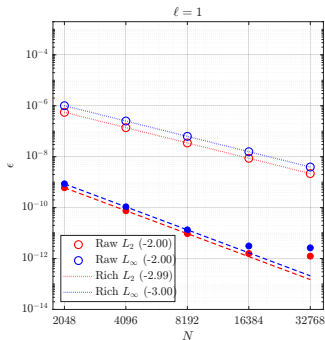


Parity error on $\mathcal{I}^+$:

$$\epsilon = \|\mathcal{R}_- - (-1)^\ell \mathcal{R}_+\|.$$

- ▶ $\ell = 1$: explicit $\mathcal{R}_+$ is **second order**; implicit $\mathcal{R}_+$ drops to **first order**.
- ▶ $\ell \geq 2$: implicit $\mathcal{R}_+$ converges at **first order**.

Richardson extrapolation: remedy for the order



Richardson extrapolation: to offset the error at leading order $O(\Delta^p)$

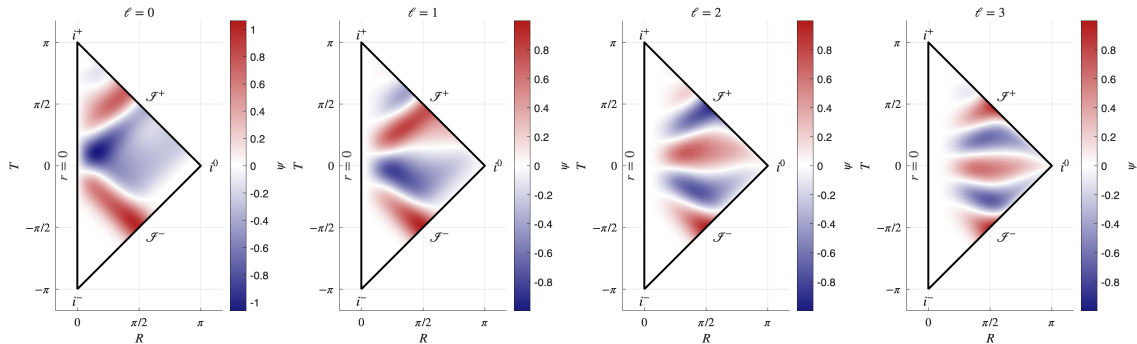
$$\mathcal{R}_{+,N}^{\text{rich},p} = \frac{2^p \mathcal{R}_{+,N} - \mathcal{R}_{+,N/2}}{2^p - 1}.$$

- $\ell = 1$ ($p = 2$): 2.00 $\rightarrow$ **3.00**.
- $\ell \geq 2$ ($p = 1$): 1.00 $\rightarrow$ **1.50** only for L_2 norm.

$$L_2 = \left(\frac{1}{N} \sum_j \epsilon_j^2 \right)^{1/2}: \text{rms of the pointwise error on } \mathcal{I}^+;$$

$$L_\infty = \max_j \epsilon_j: \text{the worst point. } N = 2\pi/\Delta.$$

Scattering off the PT potential



The original and extrapolated convergence orders are the same as those of the free case ($\ell \geq 1$).

In addition, the spherical case $\ell = 0$ is globally second-order convergent.

Scattering matrix from the time domain

Define the transfer function

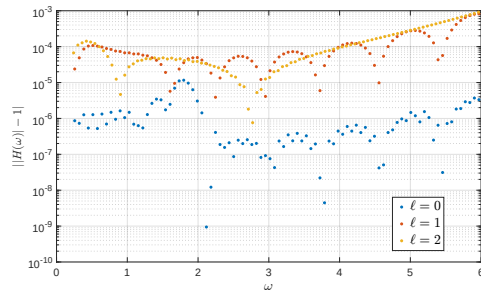
$$H_\ell(\omega) = \frac{\tilde{\mathcal{R}}_+(\omega)}{\tilde{\mathcal{R}}_-(\omega)}$$

with $\tilde{\mathcal{R}}_\pm$ the Fourier transforms of $\mathcal{R}_\pm$.

- Spectrum of the modulated Gaussian

$$\tilde{\mathcal{R}}_-(\omega) = A_0 \sigma_w \exp\left[-\frac{1}{2} \sigma_w^2 (\omega - \omega_0)^2\right]$$

- **Map** $u = \tan(U/2) \rightarrow$ **cut off** ($\delta u < \text{tol}$) $\rightarrow$ **interpolate** $\rightarrow$ FFT $\rightarrow \tilde{\mathcal{R}}_+(\omega)$
- The balance law (no transimission): $|H_\ell| = 1$

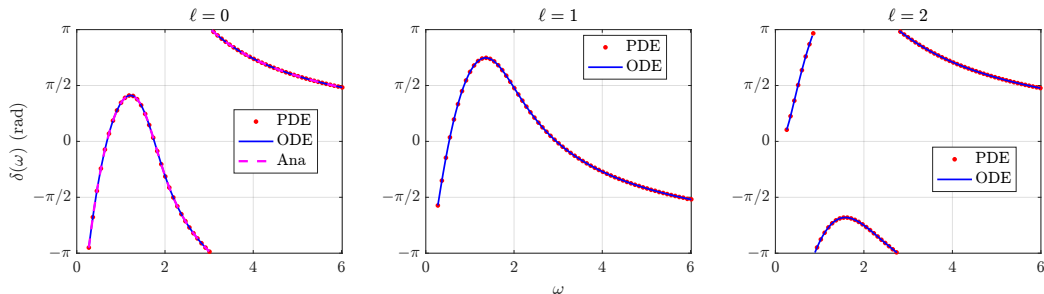


Residual deviations are below 10^{-3} .

Phase shifts: time domain vs. frequency domain

Cross-check of phase shift ($\delta_\ell = \arg H_\ell$) against the radial ODE for $\psi(t, r) = e^{-i\omega t} \psi_\omega(r)$ in frequency domain

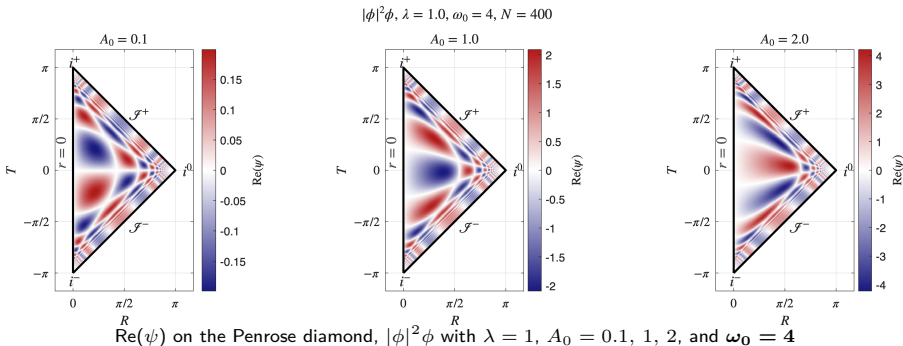
$$\left[\partial_r^2 + \omega^2 - \frac{\ell(\ell+1)}{r^2} - V_0 \operatorname{sech}^2 \left(\frac{r}{\sigma} \right) \right] \psi_\omega = 0$$



$\ell = 0$ — analytic PT S -matrix (dashed); general ℓ — Chebyshev pseudospectral solver (lines); phase shifts extracted from PDE (dots).
The agreement between all methods is $\lesssim 10^{-3}$ rad.

Semilinear $|\phi|^{n-1}\phi$: $n = 3, 5, 7$

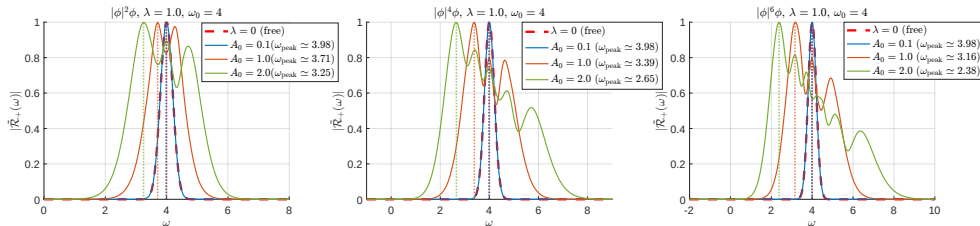
- ▶ The explicit stencil keeps $O(\Delta^2)$ globally for $\ell = 0$. Residual of energy conservation $|E_{\text{in}} - E_{\text{out}}| \sim 10^{-5}$.
- ▶ Self-defocusing Kerr effect¹: spectral broadening and multi-peak structure.



¹Intensity-dependent refractive index, a well-known phenomenon in nonlinear optics.

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spectra $|\tilde{\mathcal{R}}_+(\omega)|$ with $A_0 = 0.1, 1.0, 2.0$ vs the free wave (red dashed)

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Outline

Intro to numerical conformal scattering

Scalar wave equation in Minkowski

Regge-Wheeler equation in Schwarzschild

3.1 Schwarzschild diamond

3.2 Scattering solutions

3.3 Convergence

3.4 Transmission and reflection

Summary & outlook

From Minkowski triangle to Schwarzschild diamond

- ▶ Same compactification as Minkowski, $r \rightarrow r_*$:

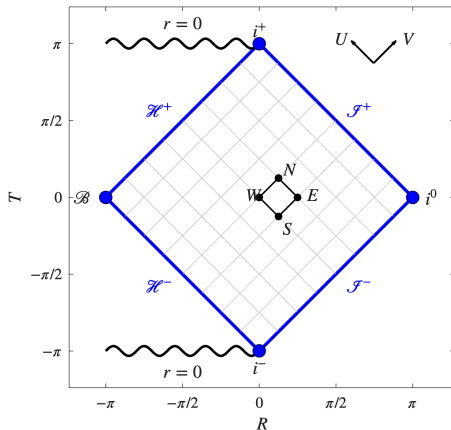
$$U = 2 \arctan(t - r_*), \quad V = 2 \arctan(t + r_*).$$

- ▶ $\Omega_P = 2 \cos \frac{U}{2} \cos \frac{V}{2} = 0$ gives the new conformal boundaries $\mathcal{H}^\pm$ and $\mathcal{B}$.
- ▶ The RW equation is recast as

$$\partial_U \partial_V \psi = -V_{\text{eff}} \psi = -\frac{1 - 2M/r}{4\Omega_P^2} \left[\frac{\ell(\ell+1)}{r^2} + (1-s^2) \frac{2M}{r^3} \right] \psi$$

with the rescaling $\psi = r\phi$ and spin $s = 0, 1, 2$.

- ▶ Goursat data: $\bullet \psi|_{\mathcal{I}^-} = \psi_{\text{inc}},$
 $\bullet \psi|_{\mathcal{H}^-} = \psi|_{\mathcal{B}} = \psi|_{i^-} = 0$
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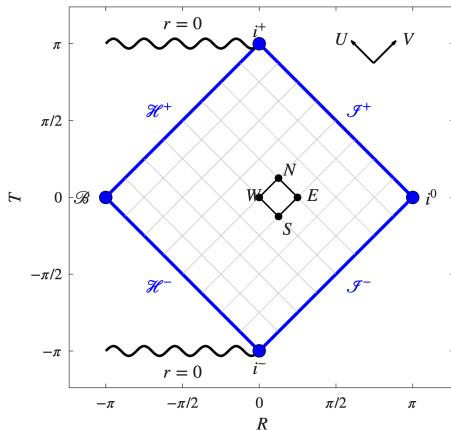
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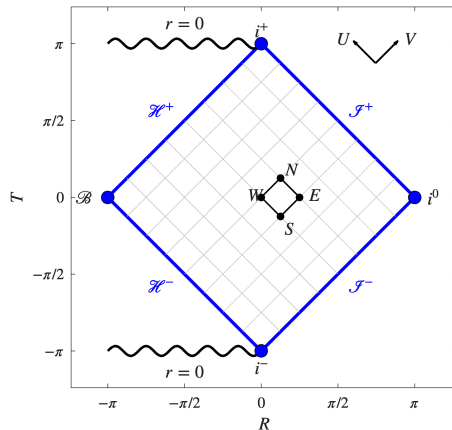
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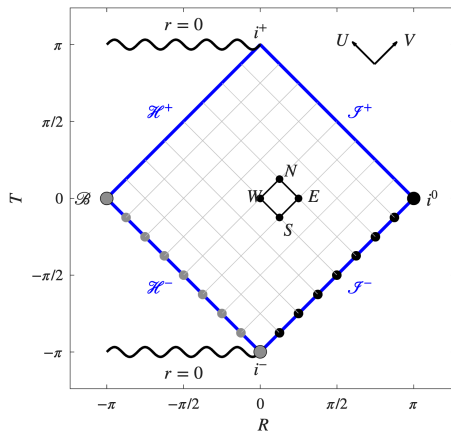
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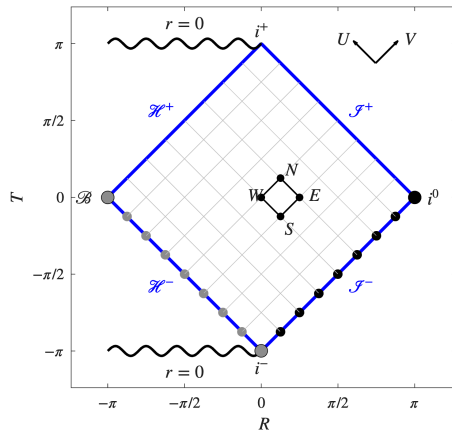
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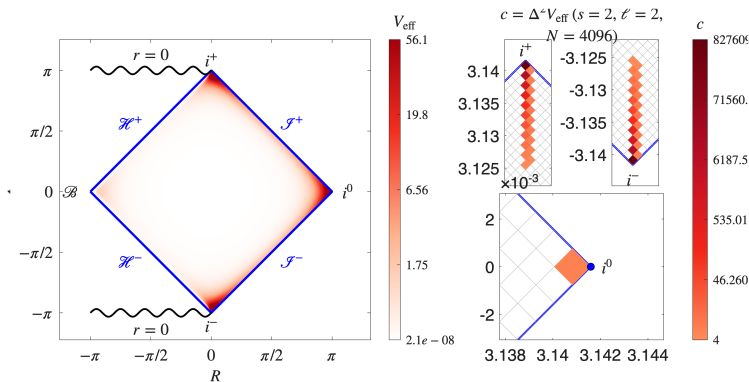


The effective RW potential at the conformal boundary

The effective RW potential

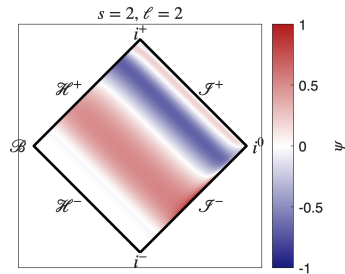
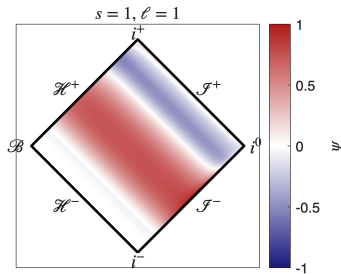
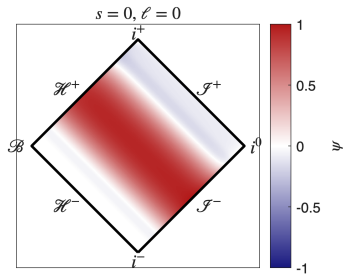
$$V_{\text{eff}} = \frac{V^{\text{RW}}}{4\Omega_{\text{P}}^2}$$

- ▶ regular on $\mathcal{I}^\pm$, $\mathcal{H}^\pm$ and $\mathcal{B}$
- ▶ diverges at i^0 as $\sim \Delta^{-2}$
- ▶ diverges at $i^\pm$ in a direction-dependent way; strongest singularity $\sim \Delta^{-4}$ along $R = 0$
- ▶ adaptive stencil for $\ell = 1$, implicit for $\ell \neq 1$

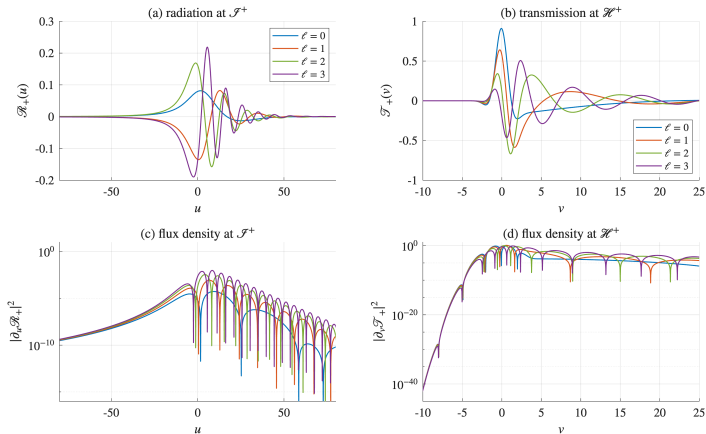


Left: V_{eff} on the diamond. Right: $c > 4$ zooms at i^+ , i^- , i^0 ($s = 2$, $\ell = 2$, $N = 4096$).

Scattering on the full diamond ($s = \ell = 0, 1, 2$)



Waveforms and flux densities ($s = 0$)



$|E_{\text{in}} - E_{\mathcal{I}^+} - E_{\mathcal{H}^+}|$ closes at $\sim 10^{-6}$ ($\ell = 0-3$).

Self-convergence ($s = 0$)

ℓ	region	Q_2	$\ e_{2N,4N}\ _2$	Q_∞	$\ e_{2N,4N}\ _\infty$
0	$\mathcal{I}^+$	2.01	4.87×10^{-7}	2.00	6.62×10^{-6}
	$\mathcal{H}^+$	2.01	4.69×10^{-7}	2.00	6.69×10^{-6}
	int	2.01	9.34×10^{-8}	2.00	6.69×10^{-6}
1	$\mathcal{I}^+$	1.98	2.98×10^{-7}	2.00	4.39×10^{-6}
	$\mathcal{H}^+$	2.00	2.25×10^{-7}	2.01	3.73×10^{-6}
	int	1.90	9.36×10^{-8}	2.00	4.37×10^{-6}
2	$\mathcal{I}^+$	0.92	1.73×10^{-4}	1.02	3.27×10^{-4}
	$\mathcal{H}^+$	2.01	1.81×10^{-6}	2.01	3.25×10^{-5}
	int	1.82	9.02×10^{-7}	1.78	1.39×10^{-4}
3	$\mathcal{I}^+$	0.82	3.85×10^{-4}	1.01	8.08×10^{-4}
	$\mathcal{H}^+$	2.02	5.51×10^{-6}	2.01	1.00×10^{-4}
	int	1.77	2.25×10^{-6}	1.69	3.61×10^{-4}

$\mathcal{I}^+$, $\mathcal{H}^+$, and interior

Grids ($N, 2N, 4N$): $\ell=0$: (2048, 4096, 8192); $\ell=1, 2, 3$: (8192, 16384, 32768)

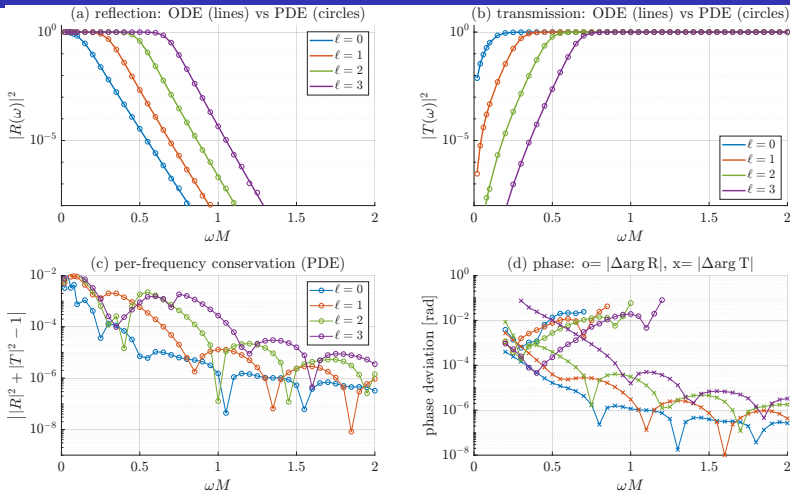
$$e_{2N,4N} := |\psi_{2N} - \psi_{4N}|$$

- ▶ elsewhere $O(\Delta^2)$ ($Q \simeq 2$)
- ▶ on $\mathcal{I}^+$, high ℓ : $Q \sim 1$
- ▶ i^+ **superconvergent**: $Q \sim 5-7$

i^+ (pointwise)

ℓ	Q_∞	$\ e_{N,2N}\ _\infty$	$\ e_{2N,4N}\ _\infty$
0	7.38	6.96×10^{-7}	4.17×10^{-9}
1	5.59	9.23×10^{-7}	1.92×10^{-8}
2	7.52	3.38×10^{-7}	1.84×10^{-9}
3	6.77	4.49×10^{-6}	4.10×10^{-8}

Check of $R(\omega)$ and $T(\omega)$



It looks okay, but still needs improvement.

Solve the stationary scattering problem in null foliations

- ▶ Introduce $z = 2M/r \in [0, 1]$; radial master function in ingoing null foliations $\psi(t, r) = e^{-i\omega(t+r_*)} \tilde{\psi}(z)$:

$$z^2(1-z)\tilde{\psi}'' + [z(2-3z) + 4i\omega M]\tilde{\psi}' - [\ell(\ell+1) + (1-s^2)z]\tilde{\psi} = 0$$

- ▶ The so-called 'in' solution ψ_{in} satisfy the scattering boundary condition. $\tilde{\psi}_{\text{in}}$ regular only in the bulk $z \in (0, 1]$.
- ▶ The 'up' branch is singular $z = 0$ in this ingoing null foliations
- ▶ $\tilde{\psi}_{\text{down}}$ regular on $[0, 1)$
- ▶ Scattering relation: $\psi_{\text{in}} = C_{\text{down}}\psi_{\text{down}} + C_{\text{up}}\psi_{\text{up}}$
- ▶ Solve $\tilde{\psi}_{\text{in}}$ on $[z_p, 1]$ and $\tilde{\psi}_{\text{down}}$ on $[0, z_p]$ with the normalization $\tilde{\psi}_{\text{in}}(1) = 1$ and $\tilde{\psi}_{\text{down}}(0) = 1$.
- ▶ Complex conjugate relation for real ω : $\tilde{\psi}_{\text{up}} = \overline{\tilde{\psi}_{\text{down}}}$
- ▶ C^1 match at $z_p \Rightarrow T = 1/C_{\text{down}}, R = C_{\text{up}}/C_{\text{down}}$

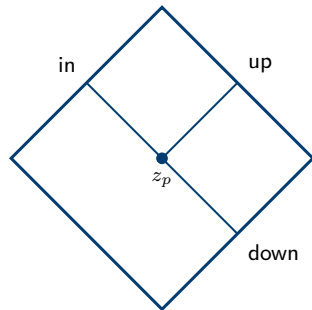


Illustration for the 'in' solution

Solve the stationary scattering problem in hyperboloidal foliations

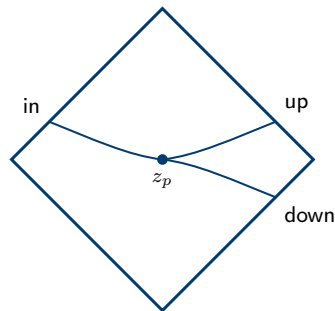
- ▶ Same $z = 2M/r$; now $\psi = e^{-i\omega(t+h)}\hat{\psi}(z)$ with height function in minimal-gauge

$$h = 2M \left[\ln \frac{1-z}{z} + 2 \ln z - \frac{1}{z} \right]$$

With $\sigma = -4iM\omega$:

$$z^2(1-z)\hat{\psi}'' + [z(2-3z) + \sigma(1-2z^2)]\hat{\psi}' - [\ell(\ell+1) + (1-s^2)z + 2\sigma z + \sigma^2(1+z)]\hat{\psi} = 0$$

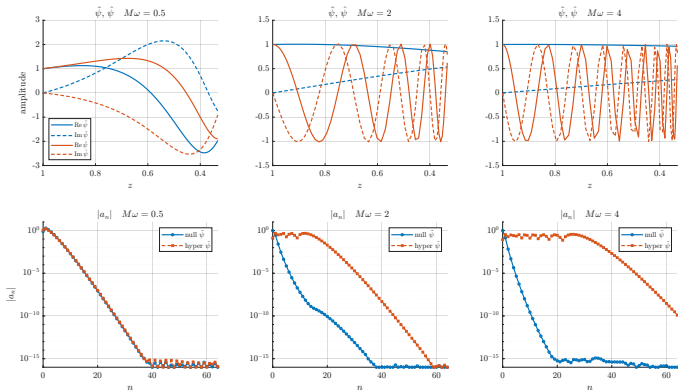
- ▶ $\hat{\psi}_{\text{in}}$ regular on $(0, 1]$, $\hat{\psi}_{\text{down}}$ on $[0, 1] \Rightarrow$ Chebyshev on $[z_p, 1]$ and $[0, z_p]$
- ▶ Real ω : $\hat{\psi}_{\text{up}} = \overline{\hat{\psi}_{\text{down}}}$, $\hat{\psi}_{\text{out}} = \overline{\hat{\psi}_{\text{in}}}$
- ▶ Same C^1 match / S -matrix at $z_p \Rightarrow R(\omega), T(\omega)$



The 'in' solution in hyperboloidal foliations

3.4 Transmission and reflection

Null vs. hyperboloidal



Top: $\tilde{\psi}_{\text{in}}$, $\hat{\psi}_{\text{in}}$ (Re solid, Im dashed; null blue, hyperboloidal orange).
 Bottom: spectral coefficients $|a_n|$ ($z_p = 1/3$, $N = 64$; $M\omega = 0.5, 2, 4$)

- Moderate ω : two foliations make no difference
- Larger ω : null converges *faster*; hyperboloidal converges *slower*.
- High-frequency waves are nearly unaffected by the potential, with $\tilde{\psi}$ almost constant.
- Hyperboloidal $e^{i\omega h(r)}$ fails to capture the bulk oscillation $e^{i\omega r_*}$
 $\Rightarrow$ more intense oscillation far away the boundary

Outline

Intro to numerical conformal scattering

Scalar wave equation in Minkowski

Regge-Wheeler equation in Schwarzschild

Summary & outlook

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Take-home message

Null foliations are a promising approach to the global evolution of conformal scattering.

Outlook

1. i^0 regularization $\rightarrow$ to increase the convergence order on $\mathcal{I}^+$
2. nonlinear scattering in less symmetric cases
3. fully nonlinear scattering
4. ...

Summary & outlook

Take-home message

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Thanks for your attention!

References

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